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Introduction to Artificial Intelligence (SE 444)

Lecture 13: Multi-Player Games and Nash Equilibrium

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A Brief History: Milestones in Game Theory

From its foundational texts to its modern applications in AI, game theory has evolved significantly.

1944: Founding Text

John von Neumann and Oskar Morgenstern publish "Theory of Games and Economic Behavior," establishing the field.

1950: Nash Equilibrium

John Nash proves that every finite game has an equilibrium point, revolutionizing strategic thinking.

1994: Nobel Prize

Nash, Harsanyi, and Selten are honored for their groundbreaking contributions to game theory.

Today: AI & Multi-Agent Systems

Game theory is fundamental for advanced AI, reinforcement learning, and autonomous systems.

The Key Insight

"I think that you think that I think..."

Game theory formalizes this recursive reasoning about what rational agents will do, capturing the essence of strategic decision-making in multi-player scenarios.

Classifying Strategic Game Environments

Strategic games possess distinct characteristics when viewed through the lens of standard AI environment properties.



Partially Observable

Opponent's exact moves are unknown until played, despite clear game structure.



Strategic Outcomes

Uncertainty stems from rational choices of others, not inherent randomness.



Multi-Agent

Features multiple rational decision-makers, each pursuing their own goals.



Static

Environment remains constant during decision-making (typical for one-shot games).



Discrete

Finite sets of strategies and outcomes, often representable in a matrix format.



Known

All players possess complete information about game structure and payoffs.

Where Does Game Theory Fit in AI?

Both **MDPs** and **Game Theory** deal with decision-making under uncertainty, but they address *different sources* of uncertainty. MDP handles uncertainty from the environment, while Game Theory handles uncertainty from **other rational agents**.

Aspect	MDP (Lecture 12)	Game Theory (Lecture 13)
Source of Uncertainty	Environment dynamics (stochastic transitions)	Other agents' strategic choices
Number of Agents	Single agent vs Nature/Environment	Multiple rational agents (2+ players)
Environment Behavior	Probabilistic but non-strategic	Strategic and reactive to your actions
Solution Concept	Optimal Policy (maximize expected utility)	Nash Equilibrium (no unilateral deviation)
Key Question	"What should I do given the world's probabilities?"	"What should I do given what others will do?"

Exploring Different Game Types

Games can be categorized based on how players interact, the nature of their outcomes, and the timing of their decisions.



Cooperative Games

Players can form binding agreements to achieve a common goal, focusing on coalition formation and fair payoff distribution.
Example: Business partnerships.



Non-Cooperative Games

Players act independently to maximize their own utility without binding agreements. This is our primary focus.
Example: Prisoner's Dilemma.



Zero-Sum Games

One player's gain is exactly another's loss, making them strictly competitive. The sum of payoffs is zero.
Example: Chess or Poker.



Non-Zero-Sum Games

Players can both win or both lose, reflecting more realistic scenarios with mixed motives.
Example: International trade agreements.



Simultaneous Games

Players make decisions at the same time, unaware of the other's choice until all actions are revealed.
Example: Rock-Paper-Scissors.



Sequential Games

Players move in a specific order, observing previous actions. Often represented by game trees.
Example: Chess or negotiation.

Why Study Multi-Player Games?



Markets

Many firms competing for customers and resources



Traffic Networks

Drivers choosing optimal routes simultaneously



Cybersecurity

Multiple attackers and defenders in strategic conflict



Multi-Agent AI

Autonomous systems interacting and adapting



The Central Question

How should a rational player act when outcomes depend on everyone else's decisions?

Game theory provides the mathematical framework to analyze these strategic interactions.

What Is a Strategic Game?

A **strategic game** (normal-form game) is defined by three core components:

01

Players

The set of decision-makers, denoted N

02

Strategies

Available actions for each player i, denoted S_i

03

Payoffs

Utility functions for each player i, denoted u_i



Mathematical Representation

We represent the complete game compactly as:

$$G = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$$

This notation fully describes the **strategic environment** and provides the foundation for all game-theoretic analysis.

Key Assumptions in Game Theory

Strategic game analysis rests on fundamental assumptions about players and game structure, crucial for predicting behavior.



Rationality

Players consistently choose actions to maximize their expected utility, based on clear preferences and optimal computation.



Common Knowledge of Rationality

Each player knows all other players are rational, and this understanding is universally known.



Common Knowledge of the Game

All players are fully aware of the game's complete structure: players, strategies, and payoffs.



Simultaneous Moves

Players make their strategic choices concurrently, without observing others' decisions beforehand.

Core Components of a Strategic Game

Every strategic game is formally defined by five essential elements that describe the players, their choices, the outcomes, and what they know.

Players N

The set of rational **decision-makers** in the game.

- $N = \{1, 2, ..., n\}$
- Players are indexed $i \in N$
- $n = |N|$ is the total number of players

Strategies S_i

The available actions or choices for each player i.

- $S_i = \{s_i^1, s_i^2, ..., s_i^m\}$
- Each player has their own strategy set.
- Strategies can be pure (specific action) or mixed (probability distribution over actions).

Action Profile s

A combination of strategies, one from each player, representing a specific outcome.

- $s = (s_1, s_2, ..., s_n) \in S$
- $S = S_1 \times S_2 \times ... \times S_n$ (Cartesian product of all strategy sets)
- s_{-i} denotes the strategies of all players except player i.

Payoff Function u_i

Maps each action profile to a utility value for player i.

- $u_i : S \to \mathbb{R}$
- $u_i(s)$ represents player i's utility when the profile s occurs.
- Higher values indicate greater preference (utility maximization).

Information Structure

Defines what each player knows about the game and other players' characteristics.

- **Complete Information:** All players know N, S_i, and u_i for every player.
- **Incomplete Information:** Some elements, like payoff functions or player types, are unknown to certain players.

Problem Formulation: From English to Game Theory

The most challenging part of game theory is translating a real-world scenario described in natural language into a formal game-theoretic model. This skill is essential for applying game theory to actual problems.

Step-by-Step Formulation Process



Identify the Players N

Ask: "Who are the decision-makers?" Look for actors making strategic choices. They must have their own goals/objectives and are usually explicitly mentioned in the problem.

List Available Strategies S_i

Ask: "What choices does each player have?" Enumerate all possible actions. Each player may have different options, and choices must be mutually exclusive.



Determine Outcomes s

Ask: "What happens for each combination?" List all strategy profiles: $s = (s_1, ..., s_n)$. Each profile leads to a specific outcome. Total outcomes = product of all strategy counts.



Assign Payoffs u_i

Ask: "How much does each player value each outcome?" Assign numerical utilities to each outcome. Higher values mean preferred outcomes. Use monetary values, years in prison, or preferences.

Players: The Set N

Player Set Definition

We assume there are n players, indexed as:

$$N = \{1, 2, \ldots, n\}$$

Key Assumptions

- Each player is **rational**
- Players maximize their own payoff
- Common knowledge of game structure

Examples

- Firms in a market
- Countries in trade negotiations
- Autonomous agents
- Network nodes

Strategy Sets

Each player i has a **finite set of strategies**:

$$S_i = \{s_i^1, s_i^2, \ldots, s_i^{k_i}\}$$

where k_i is the number of available strategies for player i.

The **joint strategy space** is the Cartesian product:

$$S = S_1 \times S_2 \times \cdots \times S_n$$

This contains **all possible outcomes** of the game.

Example: Strategy Space Size

3

Players

Number of decision-makers

2

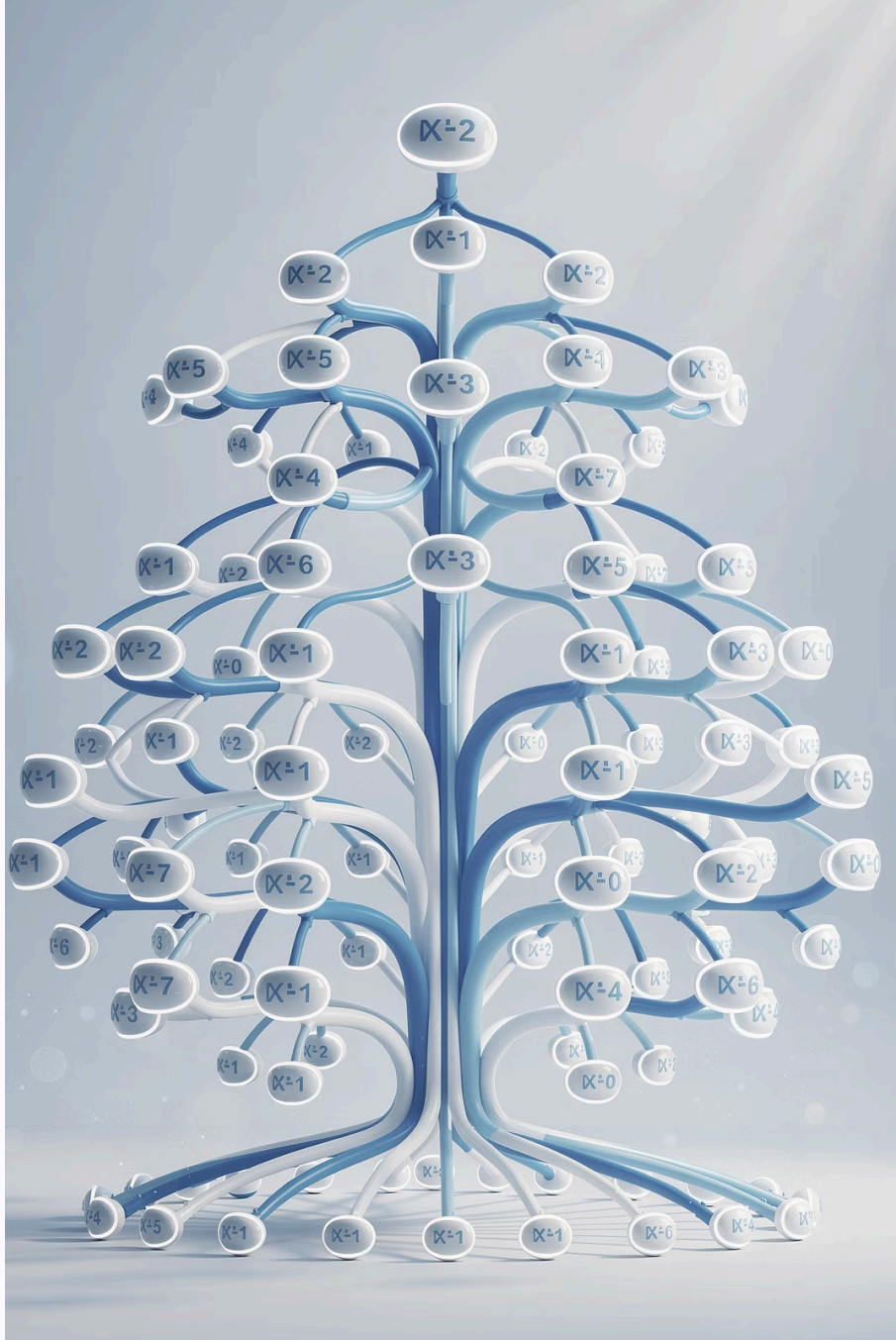
Strategies Each

Available actions per player

8

Total Outcomes

Possible strategy profiles: $2^3 = 8$



Strategy Profile

A **strategy profile** specifies one strategy for each player:

$$s = (s_1, s_2, \ldots, s_n) \in S$$

where $s_i \in S_i$ is the strategy chosen by player i.

Interpretation

- Each profile s represents one **complete outcome**
- Every player has made a decision
- The profile determines payoff for each player

Payoff Functions

Each player i has a **payoff function**:

$$u_i : S \to \mathbb{R}$$

This maps each strategy profile to a real number representing player i's utility.

What Payoffs Represent

- Money or profit
- Performance metrics
- Satisfaction or utility
- Any quantifiable preference

Key Property

Higher values are better. Different players may have **different preferences** over the same outcomes.

Separating One Player from Others

We rewrite the strategy profile using **alternative notation**:

$$s = (s_i, s_{-i})$$

1

s_i

Player i's strategy

2

s_{-i}

Strategy profile of **all other players**

where $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$

Why This Notation Matters

This notation allows us to analyze:

- What happens when **one player changes** their strategy
- While everyone else stays the same
- Essential for defining Nash equilibrium



Unilateral Deviation

A **unilateral deviation** occurs when only **one player** changes their strategy:

$$(s_i^*, s_{-i}^*) \rightarrow (s_i', s_{-i}^*)$$

where $s_i' \neq s_i^*$ and all other players keep their strategies fixed.

The Critical Question

Is this deviation profitable?

- If $u_i(s_i', s_{-i}^*) > u_i(s_i^*, s_{-i}^*)$: Player i benefits from deviating
- If $u_i(s_i', s_{-i}^*) \leq u_i(s_i^*, s_{-i}^*)$ for all s_i': No incentive to deviate

Best Response

The **best response** of player i to others' strategies s_{-i} is:

$$BR_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i})$$

Interpretation

Given what everyone else does (s_{-i}), what is the **best possible action** I can take?

Important Note

May be a **set** if there are multiple equally good responses.

Nash Equilibrium: Formal Definition

A strategy profile $s^* = (s_1^*, s_2^*, \ldots, s_n^*)$ is a **Nash Equilibrium** if:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall i \in N, \forall s_i \in S_i$$

No player can improve by unilaterally deviating. Each player is doing the best they can, given what others do.



Key Property: Strategic Stability

If players reach Nash equilibrium, no one has an incentive to change strategy

The system is in **strategic balance** – a self-enforcing state without external enforcement.



Nash Equilibrium via Best Responses

A strategy profile s^* is a Nash Equilibrium if and only if:

$$s_i^* \in BR_i(s_{-i}^*) \quad \forall i \in N$$

Interpretation

Every player is already playing a **best response** to the others. This is the **mutual best response** condition.

Practical Use

To find Nash equilibrium:

1. Compute $BR_i(s_{-i}^*)$ for each player i
2. Check if all are mutually consistent

Nash Equilibrium as Fixed Point

Define the **global best-response operator**:

$$BR(s) = BR_1(s_{-1}) \times BR_2(s_{-2}) \times \cdots \times BR_n(s_{-n})$$

Then s^* is a Nash Equilibrium if and only if:

$$s^* \in BR(s^*)$$

Mathematical Insight: Nash equilibrium is a **fixed point** of the best-response mapping, linking game theory to fixed-point theory.

Example: 3-Player Coordination Game

Setup

- **Players:** $N = \{1, 2, 3\}$
- **Strategies:** Each player chooses location A or B

Payoff Structure

All Coordinate

If **all 3** choose same location: each gets **10**

Partial Coordination

If **2** coordinate: the pair gets **7** each, lone player gets o

Sample Outcomes

1

Profile: (A, A, A)

All get 10

2

Profile: (A, A, B)

Players 1,2 get 7; Player 3 gets 0

3

Profile: (A, B, A)

Players 1,3 get 7; Player 2 gets 0



Checking Nash Equilibrium

Consider profile: (A, A, A) with current payoffs: $(10, 10, 10)$

Check each player's deviation incentive:

1 **Player 1 deviates to B**

$(B, A, A) \rightarrow$ Player 1 gets $0 < 10$ (no improvement)

2 **Player 2 deviates to B**

$(A, B, A) \rightarrow$ Player 2 gets $0 < 10$ (no improvement)

3 **Player 3 deviates to B**

$(A, A, B) \rightarrow$ Player 3 gets $0 < 10$ (no improvement)

Equilibrium Conclusion

No player benefits from deviating!

(A, A, A) is a **Nash Equilibrium**

By symmetry, (B, B, B) is also a Nash Equilibrium.

This game demonstrates **multiple equilibria** – a common feature in coordination games.

Key Properties of Nash Equilibrium



Strategic Stability

Represents a **stable state** where no unilateral deviation is profitable. Self-enforcing without external enforcement.



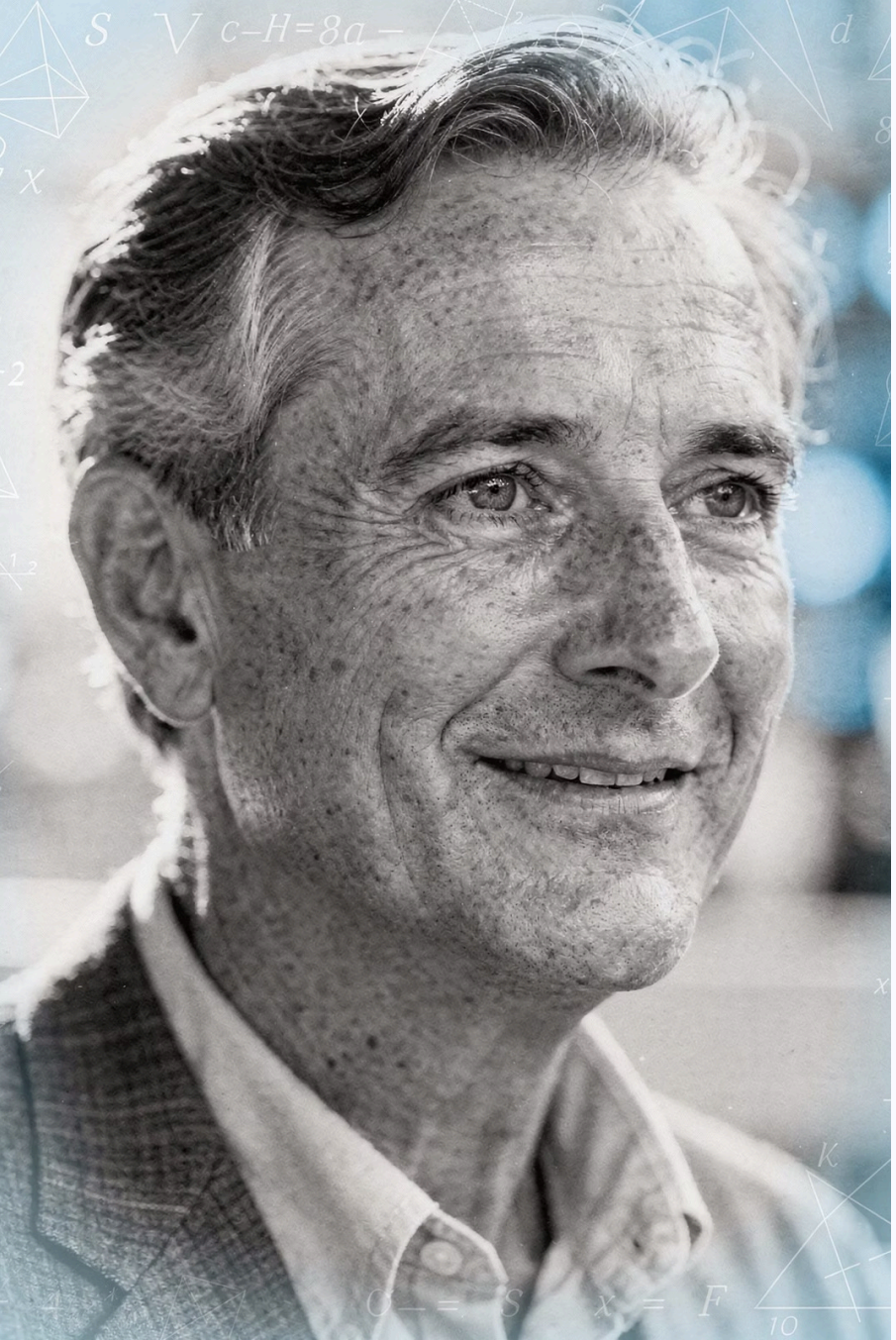
Not Always Optimal

Nash equilibrium may be **inefficient**. Individual rationality does not equal collective optimality.



Multiplicity

May be **unique** or **multiple** equilibria. Sometimes only exists in mixed strategies.



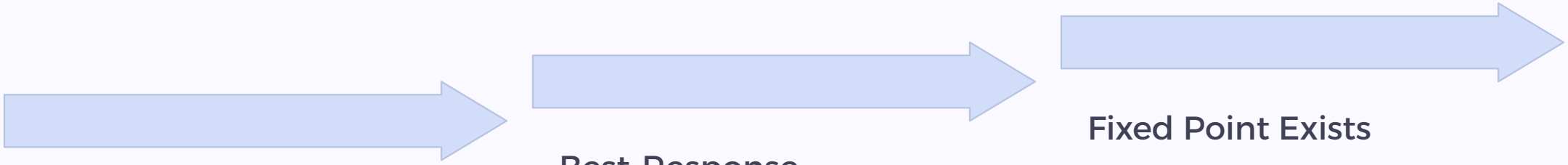
Nash Existence Theorem

> **Theorem (Nash, 1950):** Every finite n-player game has at least one Nash Equilibrium in mixed strategies.

Implications

- Strategic stability **always exists** mathematically
- May require randomization (mixed strategies)
- Pure strategy equilibrium is not guaranteed

Proof Technique



Kakutani Fixed-Point Theorem

Uses powerful mathematical tool from topology

Best-Response Correspondence

Shows that the correspondence has required properties

Fixed Point Exists

Guarantees existence of Nash equilibrium

This result is the **foundation of modern game theory**.

The Full Modeling Process

Steps to solve any multi-player game:

01

Define Players

$N = \{1, 2, \ldots, n\}$

02

Define Strategies

S_i for each player i

03

Define Payoffs

$u_i : S \to \mathbb{R}$ for each player i

04

Construct Profiles

$s = (s_1, \ldots, s_n)$

05

Compute Best Responses

$BR_i(s_{-i})$ for each player

06

Identify Equilibrium

$s^* \in BR(s^*)$

Applications Across Domains

Economics

Market competition, auctions, pricing strategies

Artificial Intelligence

Multi-agent systems, autonomous vehicles, resource allocation

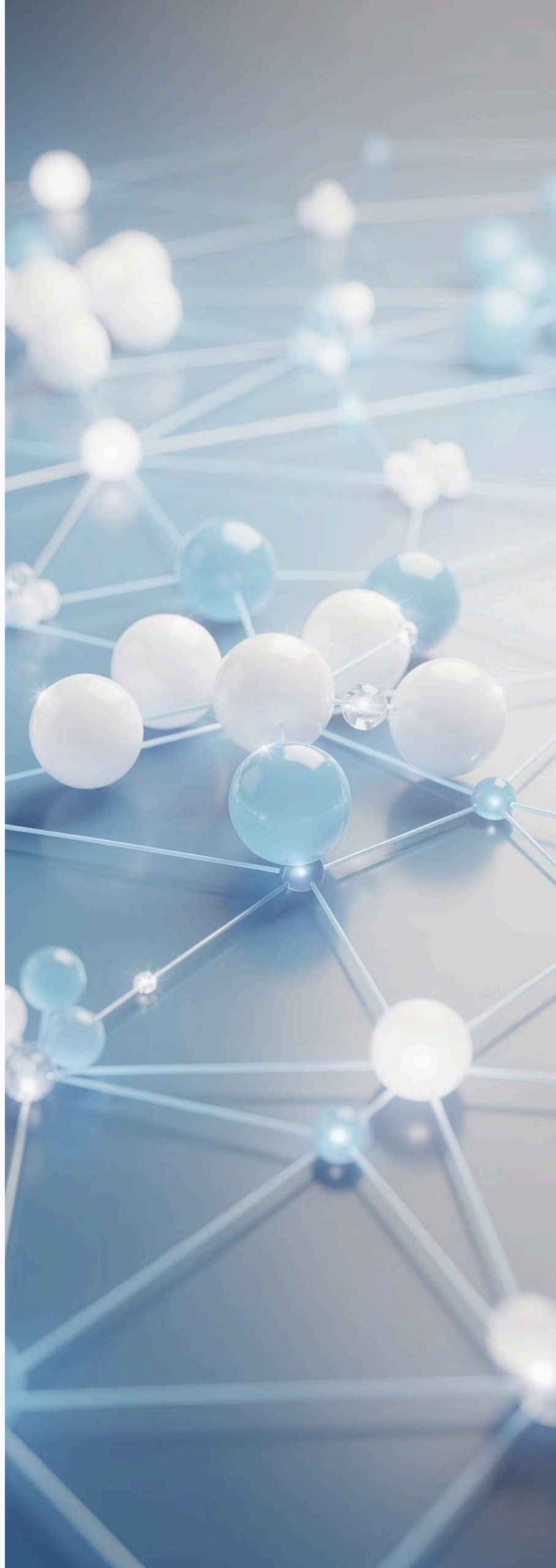
Computer Science

Network routing, cybersecurity, distributed systems

Core Insight

Nash equilibrium provides a rigorous mathematical framework for analyzing strategic interactions

It bridges theory and practice across **economics, AI, and strategic decision-making**.



Key Takeaways



Strategic Games

Defined by players, strategies, and payoffs



Best Response

Optimal action given others' choices



Nash Equilibrium

Mutual best responses – no profitable deviation



Existence

Always exists in mixed strategies



Thank You!

Questions?

Next: Mixed Strategies and Bayesian Games